

1 Projection

Logic ▾

关于 projection 的详细论述, 见 [DDA3005 Lecture 11](#), 内容包括:

- Orthogonal 和 orthogonal complement
- Orthogonal projector
- Vector decomposition
- Least square 和 orthogonal projector

1.1 Definition: Projection Matrix

我们称方阵 $H \in \mathbb{R}^{n \times n}$ 为 projection matrix, 当且仅当

1. $H^\top = H$ (symmetric)
2. $H^2 = HH = H$ (idempotent)

Remark ▾

Projection matrix 有 column space $C(H)$, 在对 H 进行 matrix multiplication 时, 可以视作 H 把其他向量 / 矩阵 project "into" 该 space

1.2 Theorem: Projection matrix 的性质

若 $H \in \mathbb{R}^{n \times n}$ 为 projection matrix, 则

- H 所有的 eigenvalues 为 0 或 1
- $\text{Tr}(H) = \sum_{i=1}^n \lambda_i = \# \text{ nonzero eigenvalues} = \text{rank}(H)$

Proof: 第一条性质的证明 ▾

由于 $H \in \mathbb{R}^{n \times n}$ 为 symmetric, 则根据 eigendecomposition, 有

$$H = PDP^\top$$

同时由于 H 为 idempotent, 可以进一步推出

$$\begin{aligned} HH &= H \\ \Rightarrow PDP^\top PDP^\top &= PDP^\top \\ \Rightarrow PDDP^\top &= PDP^\top \\ \Rightarrow DD &= D \\ \Rightarrow \lambda_i^2 &= \lambda_i, \quad \forall i \end{aligned}$$

由此得出 λ_i 为 0 或 1

| 2 Vector Calculus

| 2.1 Definition: Vector-to-Scalar Derivative

令 $f(\vec{x}) \in \mathbb{R} : \mathbb{R}^n \rightarrow \mathbb{R}$, 则其 derivative 被定义为

$$\vec{x} f(\vec{x}) = \frac{\partial f(\vec{x})}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial f(\vec{x})}{\partial x_1} \\ \frac{\partial f(\vec{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\vec{x})}{\partial x_n} \end{bmatrix} \in \mathbb{R}^n$$

⚠ Remark ▾

在本课程中, 默认使用分母布局 (denominator layout), 即 number of input = number of rows

| 2.2 Vector-to-Scalar Derivative 的例子: Linear Functions

令 $\vec{a}, \vec{x} \in \mathbb{R}^n$, 定义以下 linear function:

$$f(\vec{x}) = \vec{a}^\top \vec{x} = \sum_{i=1}^n a_i x_i$$

则其 derivative 为:

$$\frac{\partial f(\vec{x})}{\partial \vec{x}} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \vec{a} \in \mathbb{R}^n$$

| 2.3 Vector-to-Scalar Derivative 的例子: Quadratic Functions

令 $\vec{x} \in \mathbb{R}^n, A \in \mathbb{R}^{n \times n}$ 为 symmetric ($A^\top = A$), 定义以下 quadratic function:

$$f(\vec{x}) = \vec{x}^\top A \vec{x}$$

则其 derivative 为:

$$\frac{\partial f(\vec{x})}{\partial \vec{x}} = 2A\vec{x} \in \mathbb{R}^n$$

⚠ Remark: A 不 symmetric 的情况 ▾

对于 general A , 其 derivative 为:

$$\frac{\partial f(\vec{x})}{\partial \vec{x}} = (A + A^\top)\vec{x} \in \mathbb{R}^n$$

| 2.4 Definition: Vector-to-Vector Derivative

令 $f(\vec{x}) : \mathbb{R}^p \rightarrow \mathbb{R}^q$, 即

$$f(\vec{x}) = \begin{bmatrix} f_1(\vec{x}) \\ f_2(\vec{x}) \\ \vdots \\ f_q(\vec{x}) \end{bmatrix} \quad \text{here } f_i(\vec{x}) : \mathbb{R}^p \rightarrow \mathbb{R}$$

则其 derivative 为:

$$\frac{\partial f(\vec{x})}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial f_1(\vec{x})}{\partial \vec{x}} & \frac{\partial f_2(\vec{x})}{\partial \vec{x}} & \dots & \frac{\partial f_q(\vec{x})}{\partial \vec{x}} \end{bmatrix} \in \mathbb{R}^{p \times q}$$

 **Remark** 

此处仍然遵循分母布局, 即 derivative 的行数与 $\vec{x}$ 的维度相同

2.5 Vector-to-Vector Derivative 的例子: Linear Function

令 $\vec{x} \in \mathbb{R}^p, B \in \mathbb{R}^{p \times q}$, 定义以下 linear function:

$$f(\vec{x}) = B^\top \vec{x} = \begin{bmatrix} \vec{B}_1^\top \vec{x} \\ \vec{B}_2^\top \vec{x} \\ \vdots \\ \vec{B}_q^\top \vec{x} \end{bmatrix}$$

则其 derivative 为:

$$\frac{\partial f(\vec{x})}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial (\vec{B}_1^\top \vec{x})}{\partial \vec{x}} & \frac{\partial (\vec{B}_2^\top \vec{x})}{\partial \vec{x}} & \dots & \frac{\partial (\vec{B}_q^\top \vec{x})}{\partial \vec{x}} \end{bmatrix} = B \in \mathbb{R}^{p \times q} \quad (\text{注意不是 } B^\top)$$

3 Vector Probability

3.1 Definition: Random Vector

令 $X, Y \in \mathbb{R}$ 为 random variables,
则:

- X 的 distribution 为 $f_X(x)$
- Y 的 distribution 为 $f_Y(y)$
- (X, Y) 的 joint distribution 为 $f_{X,Y}(x, y) = f_{X,Y} \begin{pmatrix} x \\ y \end{pmatrix} : \mathbb{R}^2 \rightarrow \mathbb{R}$

类似的, p 个 random variables $(v_1, \dots, v_p)$ 的 joint distribution 可以写作

$$v_1, \dots, v_p \sim f \begin{pmatrix} v_1 \\ \vdots \\ v_p \end{pmatrix} : \mathbb{R}^p \rightarrow \mathbb{R}$$

或记作 $f(\vec{v})$, 其中 $\vec{v}$ 被称为 random vector

3.2 Definition: Expectation Vector

令 $\vec{v} \in \mathbb{R}^p$ 为 random vector, 则其 expectation 被定义为:

$$\mathbb{E}[\vec{v}] = \begin{bmatrix} \mathbb{E}[v_1] \\ \vdots \\ \mathbb{E}[v_p] \end{bmatrix} \in \mathbb{R}^p$$

3.3 Definition: Covariance Matrix

令 $\vec{v} \in \mathbb{R}^p$ 为 random vectors, 则其 covariance matrix 被定义为:

$$\text{Cov}(\vec{v}) = \begin{bmatrix} \text{Var}[v_1] & \text{Cov}(v_1, v_2) & \cdots & \text{Cov}(v_1, v_p) \\ \text{Cov}(v_2, v_1) & \text{Var}[v_2] & \cdots & \text{Cov}(v_2, v_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(v_p, v_1) & \text{Cov}(v_p, v_2) & \cdots & \text{Var}[v_p] \end{bmatrix} \in \mathbb{R}^{p \times p}$$

令 $\vec{u} \in \mathbb{R}^p, \vec{v} \in \mathbb{R}^q$ 为 random vectors, 则其 covariance matrix 被定义为:

$$\text{Cov}(\vec{u}, \vec{v}) = \begin{bmatrix} \text{Cov}(u_1, v_1) & \text{Cov}(u_1, v_2) & \cdots & \text{Cov}(u_1, v_q) \\ \text{Cov}(u_2, v_1) & \text{Cov}(u_2, v_2) & \cdots & \text{Cov}(u_2, v_q) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(u_p, v_1) & \text{Cov}(u_p, v_2) & \cdots & \text{Cov}(u_p, v_q) \end{bmatrix} \in \mathbb{R}^{p \times q}$$

⚠ Remark ▾

通常情况下, $\text{Cov}(\vec{u}, \vec{v}) \neq \text{Cov}(\vec{v}, \vec{u})$

3.4 Theorem: Linear Transformation

令:

- $\vec{v} \in \mathbb{R}^p, \vec{u} \in \mathbb{R}^m$ 为 random vectors

- $\vec{c} \in \mathbb{R}^q, A \in \mathbb{R}^{q \times p}, B \in \mathbb{R}^{m \times n}$ 为 constant vector 和 constant matrices 则:

- $\mathbb{E}[A\vec{v} + \vec{c}] = A\mathbb{E}[\vec{v}] + \vec{c}$

- $\text{Cov}(A\vec{v} + \vec{c}, A\vec{v} + \vec{c}) = A\text{Cov}(\vec{v})A^\top$

- $\text{Cov}(A\vec{v}, B\vec{u}) = A\text{Cov}(\vec{v}, \vec{u})B^\top \in \mathbb{R}^{q \times m}$